Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2023/2024

COURSE NAME	:	HUMAN-COMPUTER INTERACTION
COURSE CODE	:	BIM 30503
PROGRAMME CODE	:	BIM/BIP/BIS/BIW
EXAMINATION DATE	:	JANUARY/FEBRUARY 2024
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS TEST IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

THIS QUESTION PAPER CONSISTS OF **THIRTEEN (13)** PAGES

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PART A

Instruction: Answer the following questions in the answer sheet.

Q1 Ideate possibilities is one of the processes in design thinking. Which of the statements below is **FALSE**?

A. Think outside the box.
B. Identify innovative solutions to the problem statement created.
C. Tone down a wild idea to solve the problem statement created.
D. Apply brainstorming to get as many ideas as possible.

(1 mark)

Q2 Why involving the user in the 'early focus on the user and task' is one of the main principles in User Centered Design (UCD)?

A. Can get a more challenging design interface.
B. Can gain more possibilities of design outcome.
C. Can get more information and understanding of user need and their goal.
D. All the answer above is true.

(1 mark)

Q3 According to Eason, within the Higher Education System in Malaysia, who are the 'Secondary Users' that utilize the system through an intermediary?

A. Lecturers and students at UTHM.
B. Administration staff at UTHM.
C. Parents of students who study at UTHM.
D. Industries in Malaysia.

(1 mark)

Q4 Among the methods used to conduct an evaluation in any setting not involving users, which one is **NOT** typically used?

A. Inspection
B. Walkthrough models
C. Heuristics
D. Field studies

(1 mark)

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Q5 Which of the following is **NOT** one of the heuristics commonly used in a heuristic evaluation of user interface design?

A. Visibility of system status.
B. Match between the system and the real world.
C. User control and freedom.
D. Help users recognize, diagnose, and recover from errors.

(1 mark)

Q6 In the DECIDE framework for guiding evaluation, which step comes before choosing the evaluation methods?

A. Explore the questions.
B. Decide how to deal with the ethical issues.
C. Determine the goals.
D. Identify the practical issues.

(1 mark)

Q7 What is the primary purpose of a wizard in user support?

A. To replace user manuals and documentation.
B. To automate all user interactions.
C. To provide step-by-step guidance to help users complete tasks.
D. To troubleshoot technical issues.

(1 mark)

Q8 Which of the following is a current trend in user support in human-computer interaction?

A. Increase reliance on printed user manuals.
B. Limited emphasis on user feedback.
C. Utilizing AI-driven chatbots for immediate assistance.
D. Reducing the availability of customer support options.

(1 mark)

Q9 In the context of evaluation, which type of assessment is conducted during the design process to ensure that a product continues to meet users' needs?

A. Formative evaluations
B. Summative evaluations
C. Aesthetic evaluations
D. User experience evaluation

(1 mark)

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Q10 What is the primary purpose of the "help" command in DOS for user support?

A. To display a list of all available DOS commands.
B. To display a help message for a specific DOS command.
C. To create a new directory in DOS.
D. To execute a batch script.

(1 mark)

Q11 Which of the statements below is **TRUE** for the definition of user in HCI?

i. Those people who interact directly with the product.
ii. Group of people that participate in common tasks.
iii. Those people who want to achieve their task through using a specific product.
iv. People who have a purpose or aim in their mind.

A. iii.
B. i and ii.
C. i, ii and iii.
D. All the above.

(2 marks)

Q12 Who are the Stakeholders in the Higher Education System in Malaysia?

i. Lecturers and students at UTHM.
ii. Administration staff at UTHM.
iii. Parents of students who study at UTHM.
iv. Industries in Malaysia.

A. i.
B. i, ii, and iii.
C. i, ii, and iv.
D. All the above.

(2 marks)

Q13 When evaluating a system for web accessibility and design choices, where might you typically conduct the assessment to systematically investigate if all requirements are met?

A. In natural setting.
B. In a remote location.
C. In a controlled laboratory.
D. In any setting not involving users.

(2 marks)

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Q14 How Frequently Asked Questions (FAQ) can support users?

i. FAQs can serve as valuable onboarding and training resources for new users, helping them get started and learn how to use a product effectively.
ii. FAQs provide a convenient and structured way for users to access answers to common questions and issues.
iii. FAQs allow users to find answers during business hours to receive consistent and accurate information.
iv. By addressing frequently asked questions in FAQs, companies can reduce the number of repeated queries that reach their support teams, allowing support staff to focus on more complex and unique issues.

A. i, ii and iii
B. ii and iii
C. i, ii and iv
D. ii and iv

(2 marks)

Q15 What combination of media or formats would be most well-rounded and effective for enhancing user understanding in an online tutorial for Python programming for website development?

A. Video tutorials, infographics and diagrams, and code examples.
B. Interactive code editors, checklist, and mind map.
C. Augmented Reality and Virtual Reality.
D. A crossword puzzle related to Python terminology.

(2 marks)

Answer sheet for PART A

1		6		11	
2		7		12	
3		8		13	
4		9		14	
5		10		15	

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PART B

Instruction: Answer the following questions in the space provided.

Q16 Questions **Q16(a) – Q16(b)** are based on the following scenario.

Public information kiosks typically feature touchscreens with large buttons. Upon closer inspection of **Figure Q16.1**, you will notice that each button displays only one or more words, indicating the title of the next screen. This simplicity may lead users to unintentionally navigate to the wrong screen, necessitating the use of the 'back' or 'home' button to retry.

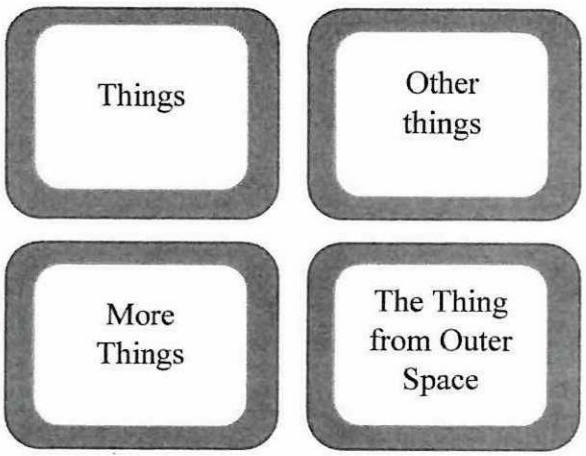


Figure Q16.1

(a) Explain **FOUR (4)** reasons why those buttons not helping fulfill user needs. Your answers also need to be supported by using first principles of User Centered Design (UCD).

(8 marks)

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(b) Redesign the buttons in Figure **Q16.1** and give a simple explanation of your new button and word design.

(12 marks)

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Q17 Imagine you are conducting a user evaluation for a new mobile application and you are responsible for ensuring the protection of users' rights.

(a) Discuss in detail **FIVE (5)** steps on how to protect users' rights. In your discussion, provide a comprehensive explanation of each step.

(10 marks)

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(b) The Association for Computing Machinery (ACM) plays a crucial role in maintaining ethical standards in the field of computing. Explain **THREE (3)** ways in which adhering to ethical codes, like those established by the ACM, enhances the ethical conduct of evaluations and research in technology-related studies.

(6 marks)

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(c) Evaluate, analyze, interpret, and present the data is the last item on the DECIDE framework to guide evaluation. Discuss **TWO (2)** key considerations when evaluating, analyzing, and presenting data in the context of Human-Computer Interaction research.

(4 marks)

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Q18 (a) Discuss **TWO (2)** differences between **Figure Q18.1** and **Figure Q18.2** as user support tools.

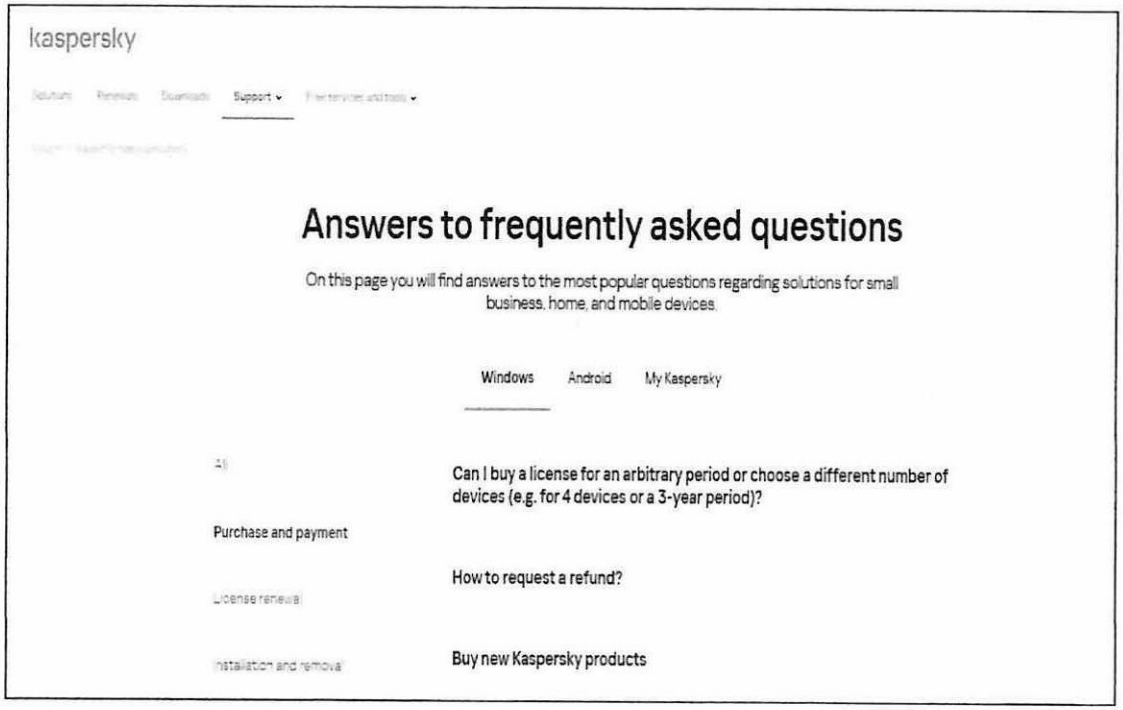


Figure Q18.1



Figure Q18.2

(10 marks)

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(b) Using *Sistem Maklumat Akademik Pelajar* (SMAP) UTHM as a case study, propose **FOUR (4)** enhancements in user support for the portal to maximize its effectiveness.

(10 marks)

- END OF QUESTIONS -

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